To investigate the differences in models A and B (pictured below), I used AI to help me develop a MATLAB script that could graph the probability of remaining in state 1 (or 0 for model B) and the probability of being in the K manifold as a function of time. This probability relies on a variety of conditions including:

N - the number of K states
ΔE - the spread of energy states for the K manifold
ρ^{-1} - the inverse of the density of states of K
V_{1K} - the coupling between state 1 and the K states
V_{01} - the coupling between state 0 and state 1
E_1 - the energy of state 1 (0 in this case)
E_0 - the energy of state 0

By varying the following conditions, I was able to better understand how this model the time evolution of this quantum system. The starting conditions I used are described below and followed the rule that $\Delta E >> V >> \rho^{-1}$. This rule is important because if the coupling is larger than the energy spread, there will just be oscillations between the two states and if the inverse of the density of states is larger than the coupling and energy spread, there will not be good interaction between state 1 and states K.

Test 1: Initial Conditions
N = 100
$\Delta E = 5$
$\rho^{-1} = 0.05$
$V_{1K} = 0.1$
$V_{01} = 0.2$
$E_1 = 0$
$E_0 = -1$

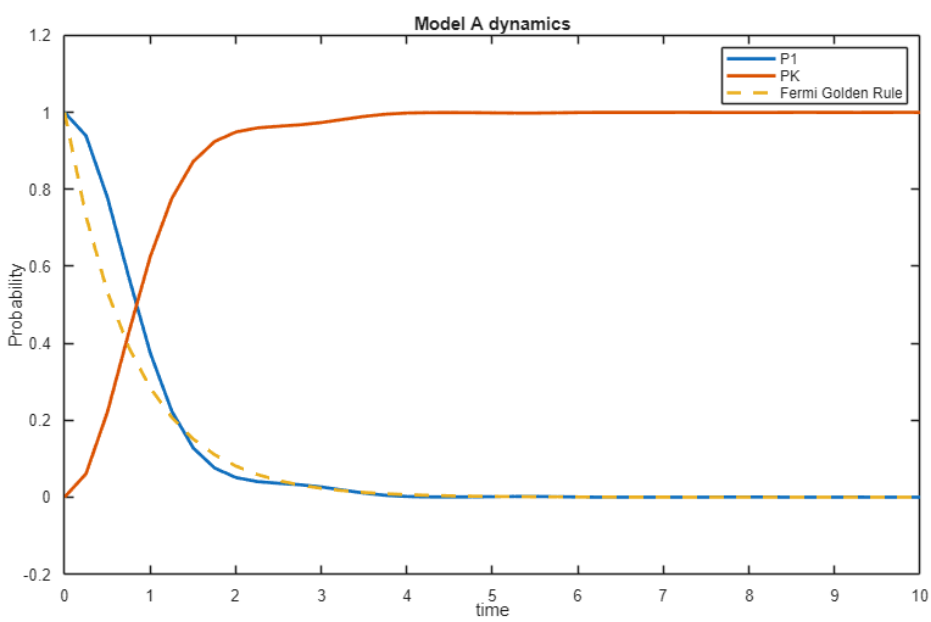

Figure 1. Model A for test 1 (initial conditions).

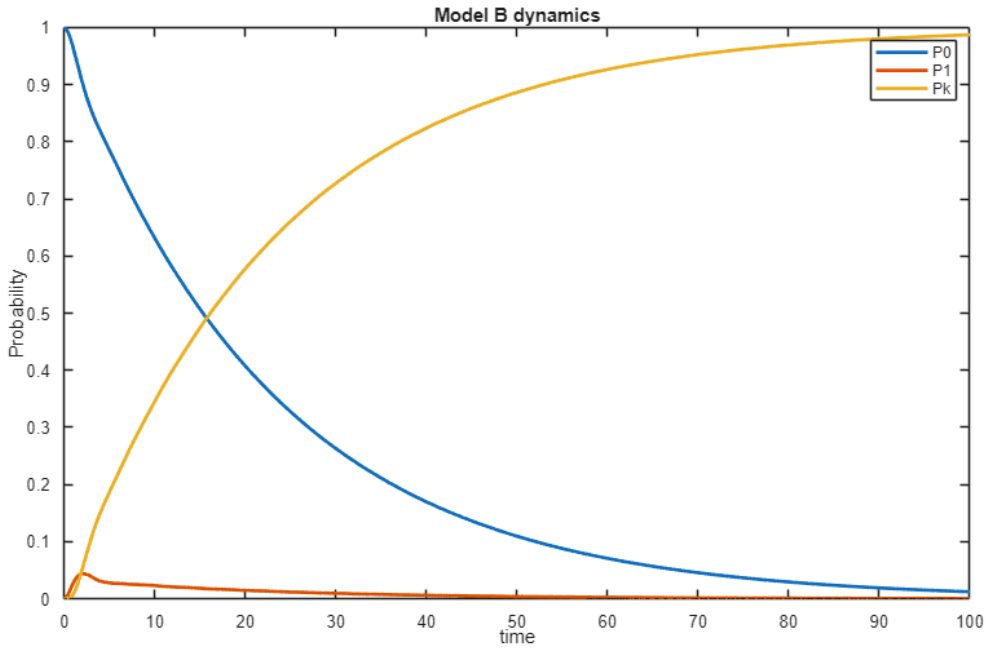

Figure 2. Model B for test 1.

The first test conditions modeled exactly what I would've expected to happen given my previous knowledge of relaxation. For model A (figure 1), the probability of being in state 1 decreases as the probability of being in state K increases until the system has completely relaxed. This model even follows closely to Fermi's golden rule which uses the assumption of infinite states for K. For model B (figure 2), the the probability of being in state 0 decreases and the probability of being in state K increases while the probability of being in state 1 increases slightly then goes back to zero, likely due to the small coupling value between state 0 and state 1. The decrease in probability of being in state 0 to zero is much slower than state 1 because the system starts in state 0 and must go up in energy to be in state 1. Then once it is in state 1, it can interact with states K. This means it takes much longer for the system to relax completely due to the extra step involved. Additionally, when the coupling value is large between state 0 and state 1, the probability of being in state 1 increases initially and then goes back to zero and the probability of being in state 0 drops much faster (figure 3). Finally, the value of E_0 - E_1 has little to no effect on the probability of being in state 1 (figures 4-5).

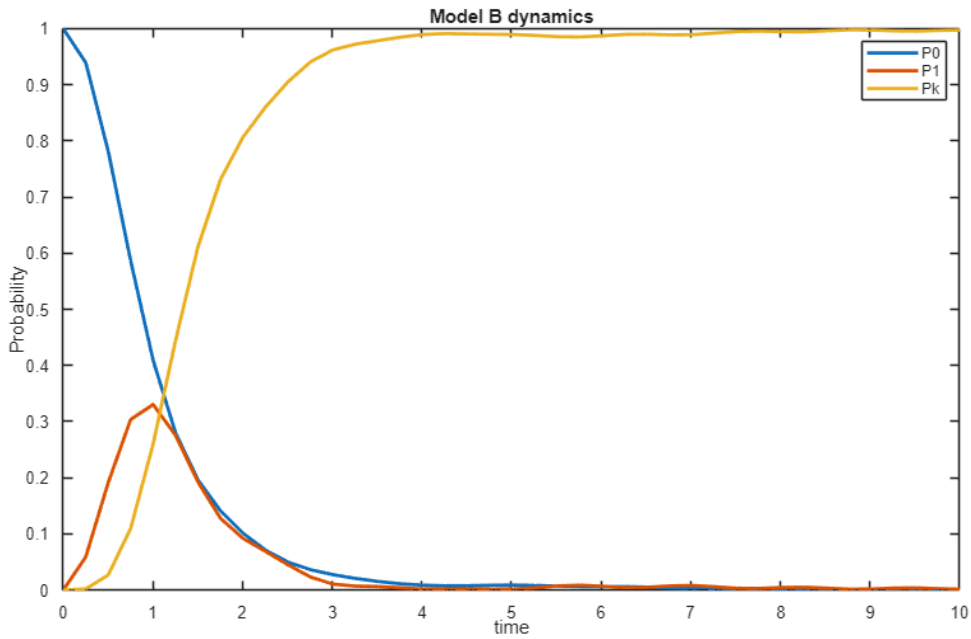

Figure 3. Model B for test 1 with V_{01} of 1 instead of 0.1.

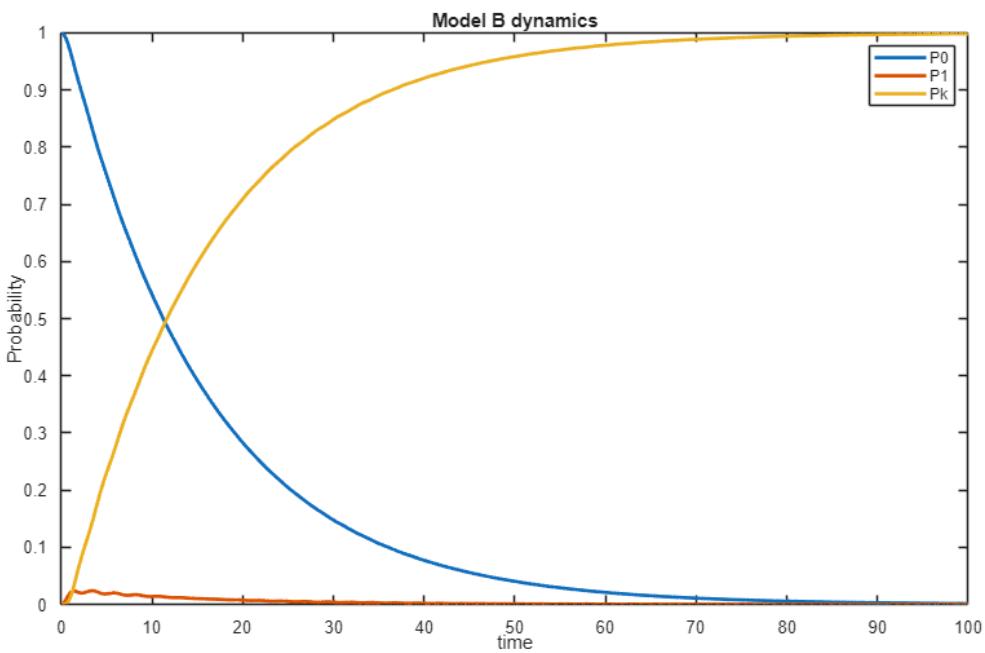

Figure 4. Model B for test 1 with E_0 of -0.1 instead of -1.

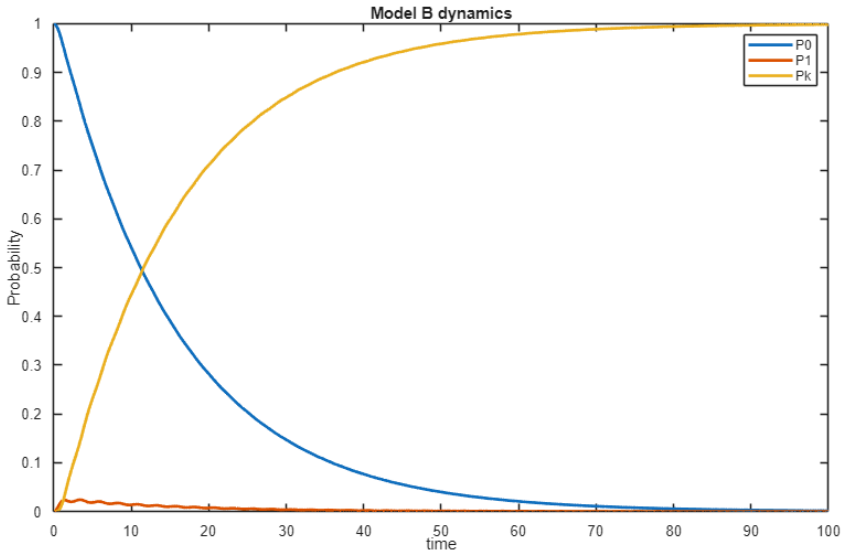

Figure 5. Model B for test 1 with E_0 of -0.01 instead of -1 or -0.1.

Test 2: Increasing N

N = 200

$\Delta E = 5$

$\rho^{-1} = 0.025$

$V_{1K} = 0.1$

$V_{01} = 0.2$

$E_1 = 0$

$E_0 = -1$

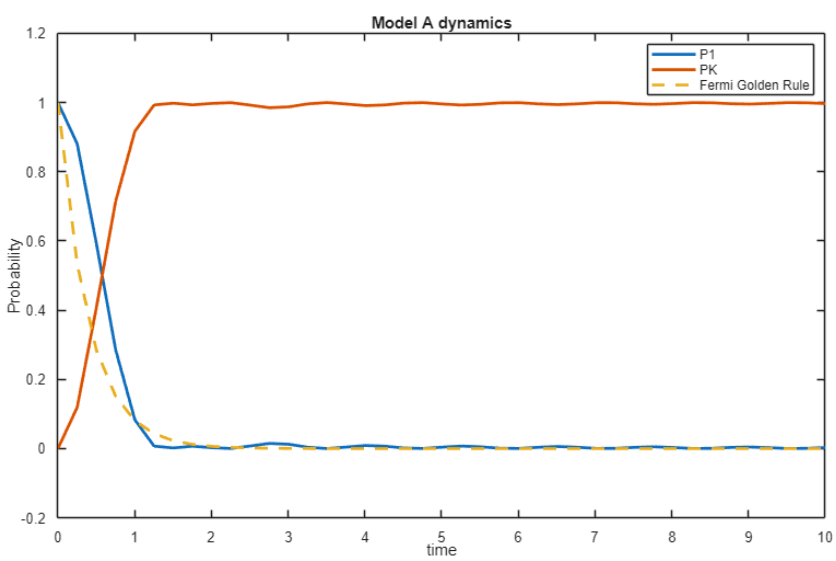

Figure 6. Model A for test 2 (increase in value of N).

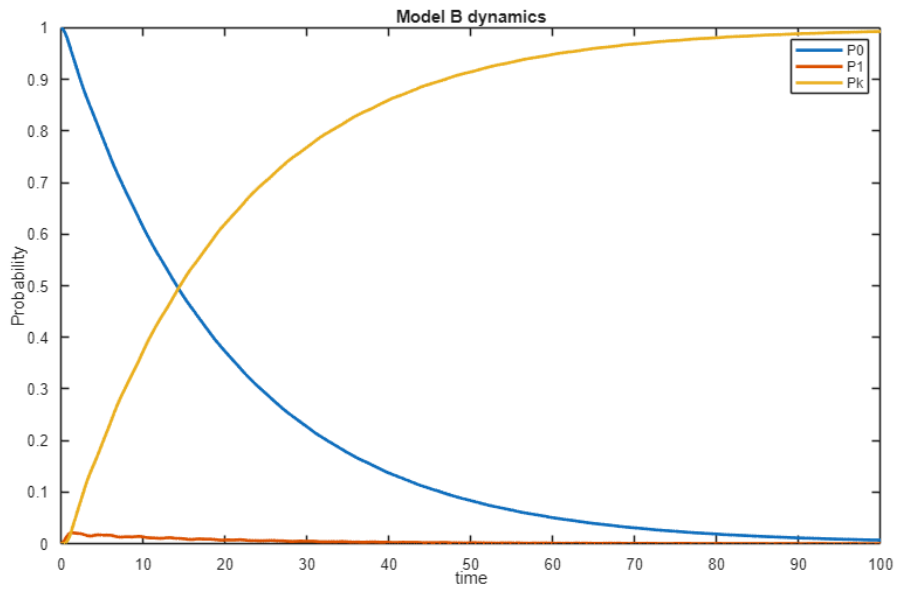

Figure 7. Model B for test 2.

The second test conditions increased the value of N from 100 to 200. The showed that increasing the value of and therefore the density of states resulted in a faster relaxation for model A (figure 6) and not much of a change for model B (figure 7). This makes sense that it resulted in faster relaxation because there were more states for state 1 to interact with. Additionally, it also makes sense that it did not have a noticeable effect on model B because the rate of relaxation is still dependent on the interaction between state 0 and state 1, which had not changed.

Test 3: Increasing ΔE

N = 200
$\Delta E = 20$
$\rho^{-1} = 0.1$
$V_{1K} = 0.5$
$V_{01} = 0.2$
$E_1 = 0$
$E_0 = -1$

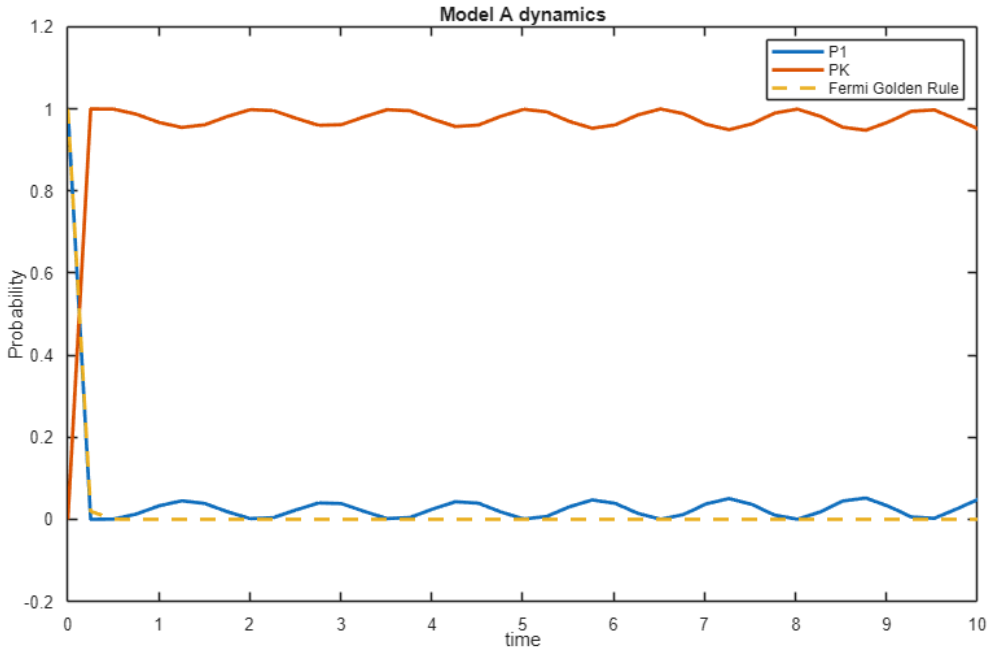

Figure 8. Model A for test 3 (increasing ΔE).

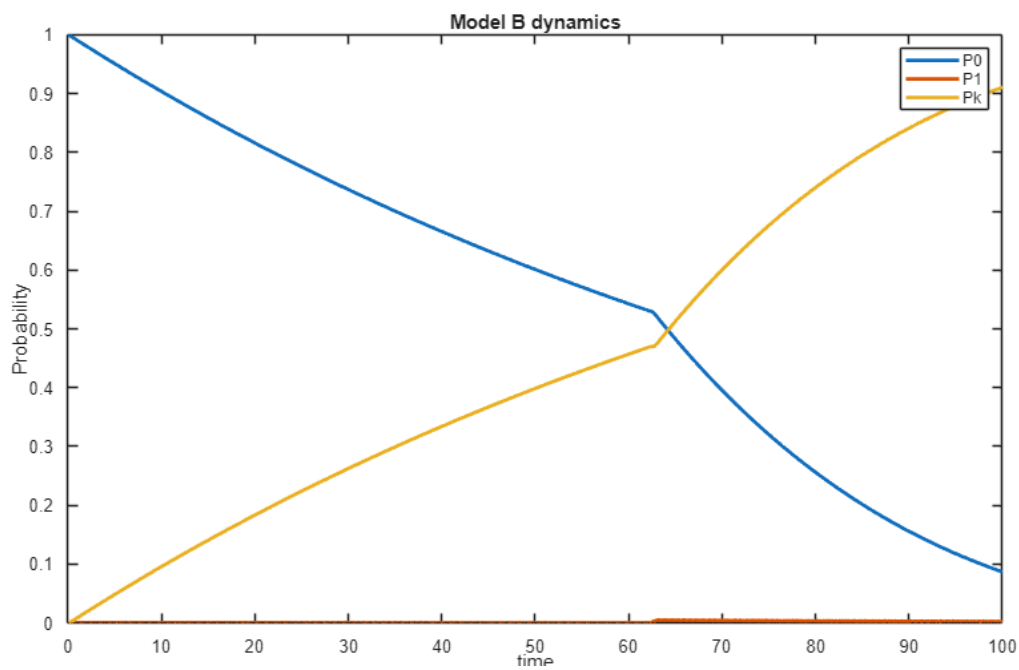

Figure 9. Model B for test 3.

For test 3, the value of ΔE was increased from 5 to 20. (The value of V was also increased from 0.1 to 0.5 to ensure ΔE >> V >> ρ^{-1} still remained true. Increasing ΔE resulted in an even faster relaxation in model A, as well as more intense oscillations between state 1 and state K around a probability of zero and one, respectively (figure 8). I'm not sure why there are oscillations resulting in negative probability and a probability above one. It is likely the way the model is set up. Additionally, in model B, there is no longer an exponential decay and growth for

the probability of being in state zero and state K, respectively (figure 9). There is also zero probability of being in state 1 which is in agreement with the increasing of the density of states. As soon as a molecule entered state one, it would likely go to states K. Given the agreement with Fermi's golden rule in model A, a larger ΔE (closer to infinity) is better for this model. However, model B does not handle this increase in density of states well.

Test 4: Increasing V

N = 200
ΔE = 10
ρ^{-1} = 0.05
V_{1K} = 1
V_{01} = 1
E_1 = 0
E_0 = -1

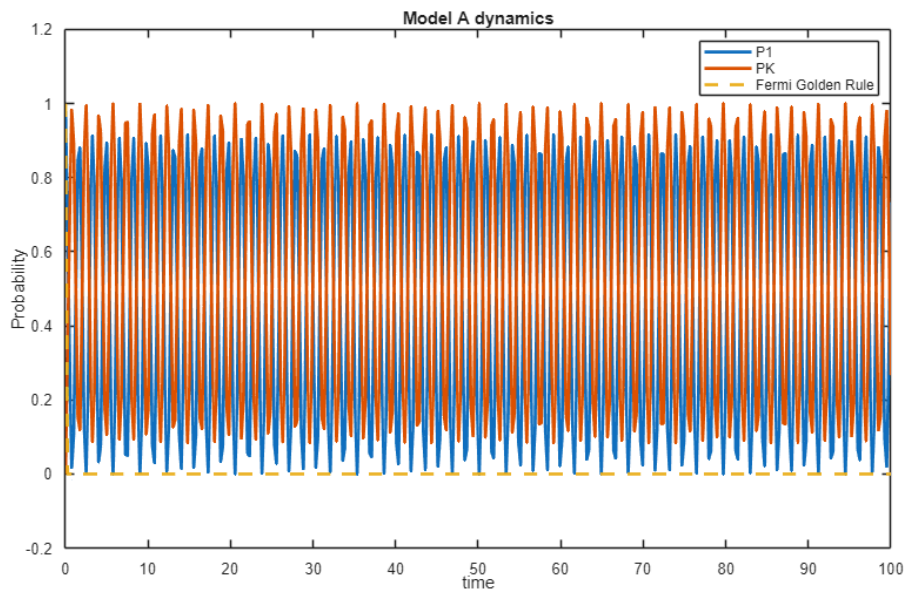

Figure 10. Model A for test 4 (increasing V).

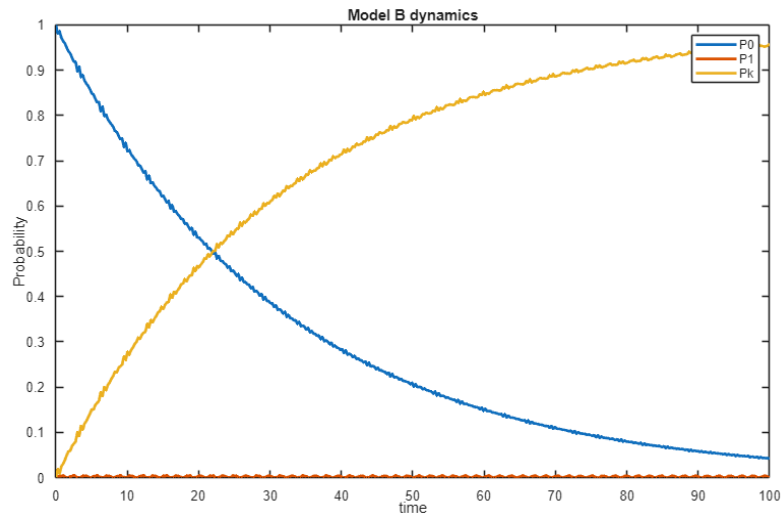

Figure 11. Model B for test 4.

For test 4, the value of V_{1K} and V_{01} was increased to 1. For both model A and model B, this resulted in an increase in oscillations throughout the entire time investigated. For model A, this resulted in a continuous oscillation between state 1 and states K as time increased (figure 10). This likely means the coupling between the two states was too strong and resulted in the oscillations between the states rather than relaxation. For model B, the relaxation still occurred, just with some oscillations as it happened (figure 11). This means the coupling between state 1 and states K is much more important in model A than in model B, which aligns with what would be expected.

Test 5: Random sign on V

Conditions:

N = 200

$\Delta E = 5$

$\rho^{-1} = 0.025$

$V_{1K} = \pm 0.1$

$V_{01} = \pm 0.2$

$E_1 = 0$

$E_0 = -1$

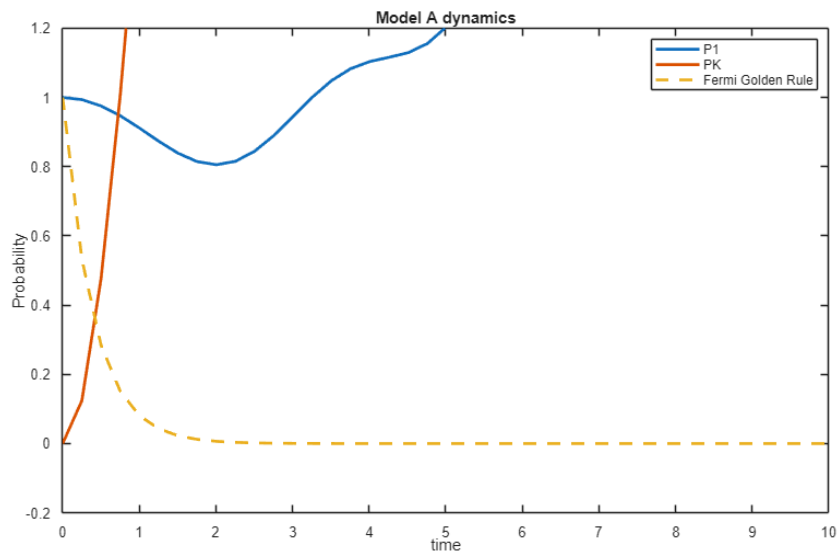

Figure 12. Model A for test 5 (random sign on V).

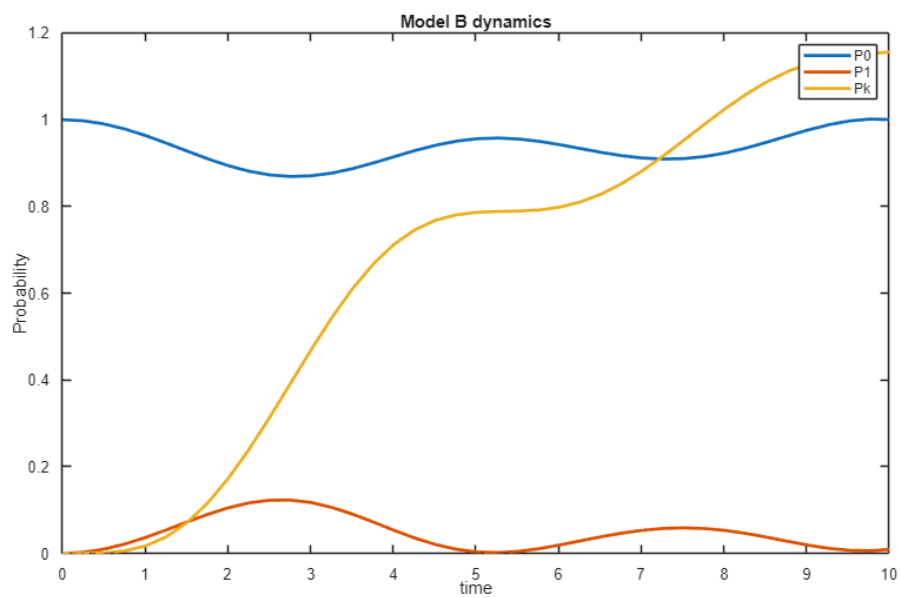

Figure 13. Model B for test 5.

For test 5, I randomly assigned a positive or negative sign to V. This completely broke my model, resulting in very large probabilities at long time scales. At the shorter time scale picture above (figures 12-13), model A was nowhere close to what the relaxation should look like and while model B was closer, it did not hold up at larger time scales (figure 14). I'm not sure why this happened, given that sign should not affect probability which contains an absolute value squared term.

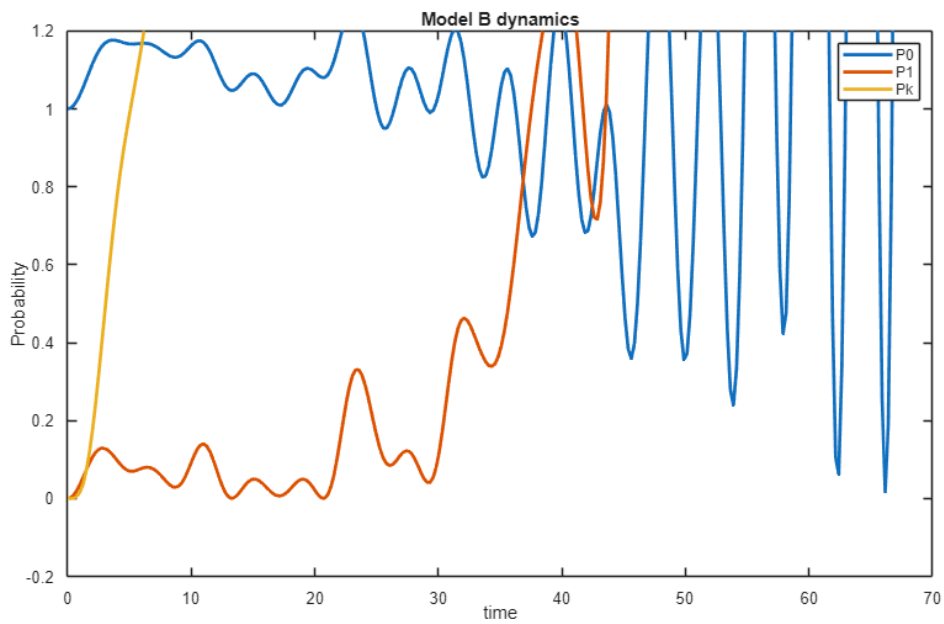

Figure 14. Model B for test 5 at longer time scale.

Test 6: Breaking ΔE >> V >> ρ^{-1} rule

N = 200

ΔE = 40

ρ^{-1} = 0.2

V_{1K} = 0.1

V_{01} = 0.2

E_1 = 0

E_0 = -1

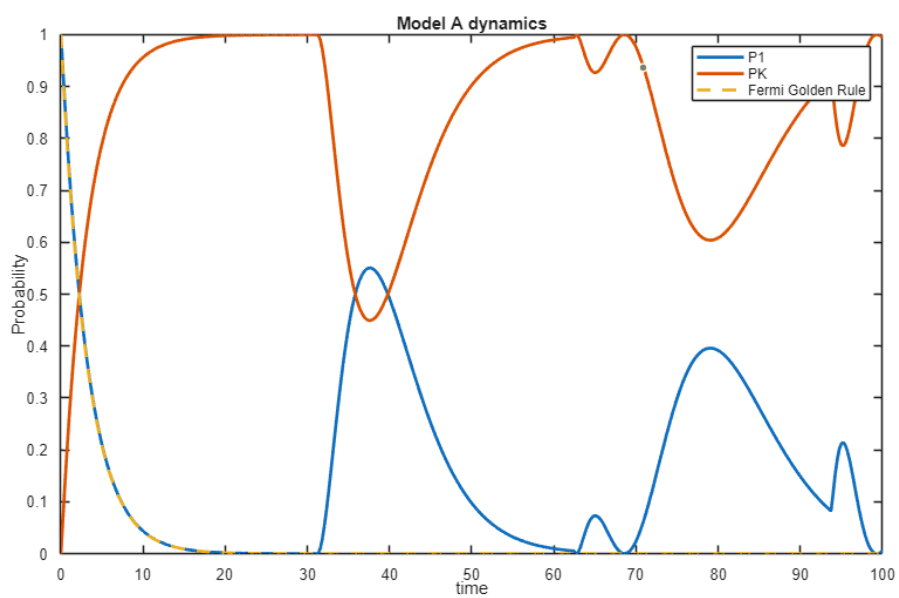

Figure 15. Model A for test 6 (breaking ΔE >> V >> ρ^{-1} rule).

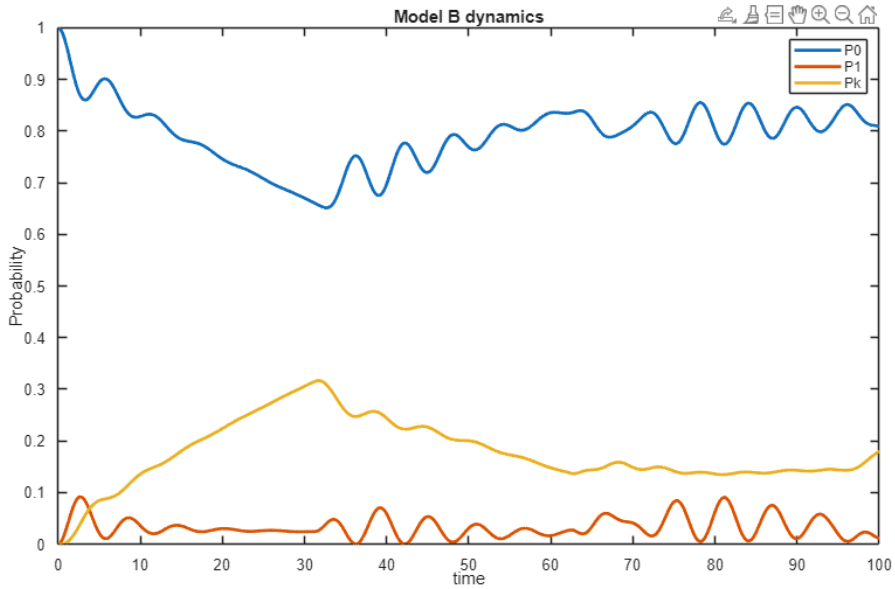

Figure 16. Model B for test 6.

For test 6, I intentionally broke the rule ΔE >> V >> ρ^{-1} and made ρ^{-1} larger than V. For model A, this resulted in a deviation from the expected relaxation at large time scales (figure 15). At small time scales (<20 sec), the model functions as expected but then it starts to oscillate in strange patterns. For model B, the model breaks down with strange oscillations and no real identifiable pattern. This shows that this rule is important for maintaining a good model for relaxation.

Test 7: Random distribution of V with average of zero

N = 100

ΔE = 5

ρ^{-1} = 0.2

V_{1K} = <V> ~ 0 and $\langle V^2\rangle^{1/2} \ll \Delta E$

V_{01} = <V> ~ 0 and $\langle V^2\rangle^{1/2} \ll \Delta E$

E_1 = 0

E_0 = -1

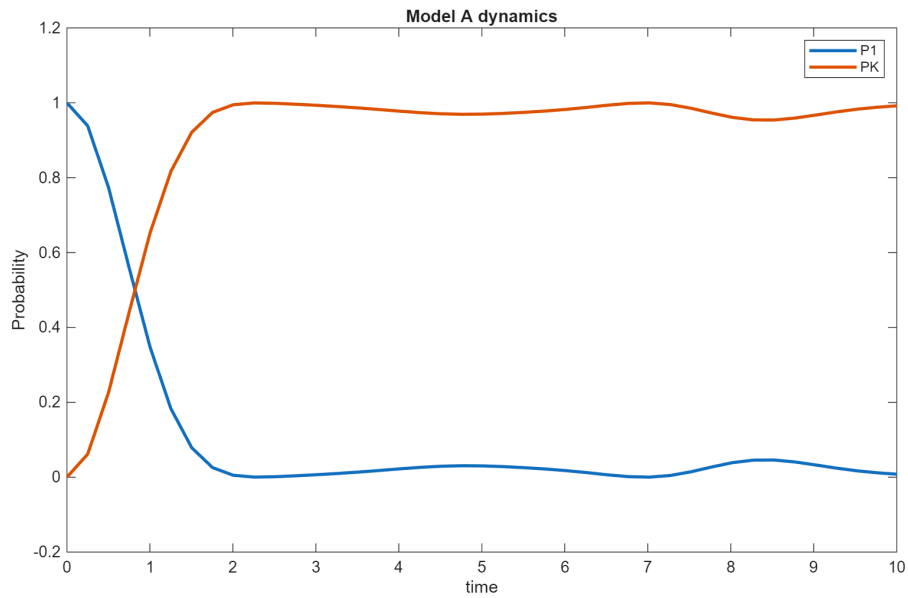

Figure 17. Model A for test 7 (random distribution of V).

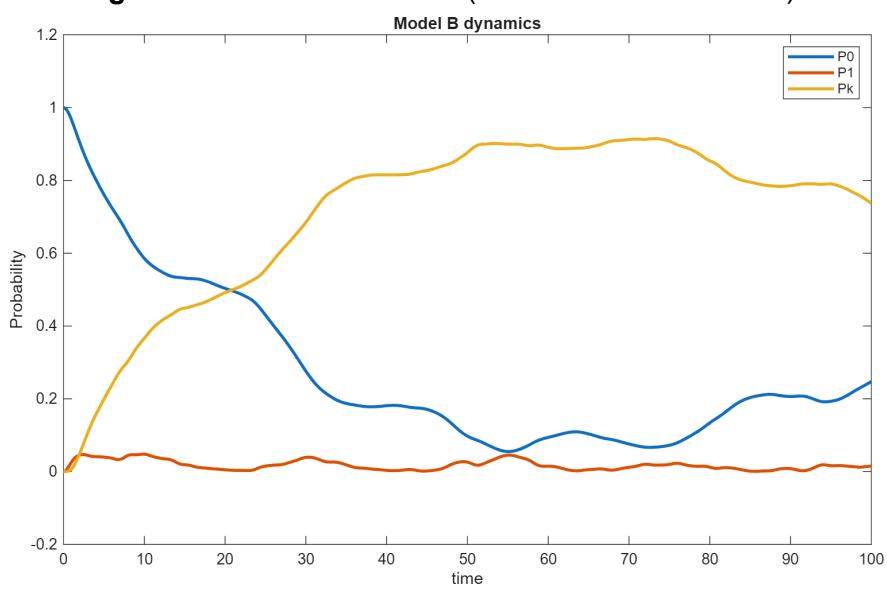

Figure 18. Model B for test 7.

Finally, for test 7, I had a random distribution of V with an average of approximately zero and a RMS less than the spread of energy and the inverse of the density of states. For model A, there is still a good approximation of exponential decay/growth, similar to a V of 0.1 (figure 17). For model B, there are some nonuniform oscillations but there is still the exponential decay/growth (figure 18). This is not exact because I could not figure out the code to make the mean exactly zero. The mean values of the graphs above were -0.002 and 0.0028, respectively.

In conclusion, I found that both models were pretty robust for a variety of different conditions. Changing the conditions resulted in different types of exponential changes but almost always gave exponential changes.